Experiment:- 6

Problem statement:-

Linux operating systems support multiple user sessions and interfaces through the use of virtual terminals and graphical environments. Understanding how Linux manages text based terminals, graphical display servers, display managers, X clients, and window managers is essential for system administration, troubleshooting, and cyber security operations.

The problem is to explore and understand virtual terminals, display managers, X clients, and window managers, and to practice switching between text-based and graphical interfaces in a Linux environment.

Aim:- Understanding virtual terminal, display manager, X clients, and window manager in linux.

Procedure:-

1. Log in to the Linux system and open the terminal. Log in to the Linux system. Open the terminal. Check the current terminal: tty

2. Switch to another virtual terminal using: Ctrl + Alt + F2. Log in and verify terminal access. Switch back to graphical mode: Ctrl + Alt + F1 or Ctrl + Alt + F7

3. Check logged-in users: who w

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ who
srishti_m seat0         2026-03-28 02:12
srishti_m tty1          2026-03-28 02:12
srishti_m seat0         2026-03-28 01:30 (:0)
srishti_m seat0         2026-03-28 02:10
srishti_m tty2          2026-03-28 02:10
```

4. Check the display manager status: systemctl status lightdm

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ systemctl status lightdm
● lightdm.service - Light Display Manager
     Loaded: loaded (/usr/lib/systemd/system/lightdm.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-03-28 01:30:11 EDT; 45min ago
 Invocation: df6757d183334611bd018db6557e3748
       Docs: man:lightdm(1)
   Main PID: 782 (lightdm)
      Tasks: 6 (limit: 4264)
     Memory: 119.5M (peak: 126.8M)
        CPU: 24.936s
     CGroup: /system.slice/lightdm.service
             ├─782 /usr/sbin/lightdm
             └─794 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten >

Warning: some journal files were not opened due to insufficient permissions.
lines 1-14/14 (END)
```

5. Verify X session variable: echo $DISPLAY

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ echo $DISPLAY
:0.0

┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$
```

6.List running X-related processes: ps -e | grep X

```
┌──(srishti_m㉿kali)-[~/grep_lab1]
└─$ ps -e | grep x
     19 ?        00:00:00 rcu_exp_par_gp_kthread_worker/0
     20 ?        00:00:00 rcu_exp_gp_kthread_worker
    307 ?        00:00:00 irq/18-vmwgfx
    361 ?        00:00:00 kworker/R-ext4-rsv-conversion
    648 ?        00:00:00 kworker/R-xprtiod
    713 ?        00:00:00 VBoxService
   1048 ?        00:00:00 mpris-proxy
   1064 ?        00:00:00 xfce4-session
   1130 ?        00:00:00 VBoxClient
   1131 ?        00:00:00 VBoxClient
   1145 ?        00:00:00 VBoxClient
   1147 ?        00:00:01 VBoxClient
   1153 ?        00:00:00 VBoxClient
   1154 ?        00:00:08 VBoxClient
   1225 ?        00:00:00 VBoxClient
   1226 ?        00:00:00 VBoxClient
   1232 ?        00:00:10 xfwm4
   1262 ?        00:00:01 xfsettingsd
   1268 ?        00:00:01 xfce4-panel
   1282 ?        00:00:02 xfdesktop
   1333 ?        00:00:00 xfce4-notifyd
   1359 ?        00:00:01 xfce4-screensav
   1365 ?        00:00:00 xiccd
   1376 ?        00:00:00 xfce4-power-man
   1397 ?        00:00:00 xcape
```

Expected Output

- Successful switching between virtual terminals.
- Identification of display manager and X session.
- Display of active user sessions.
- Confirmation of graphical environment variables.

Result

Virtual terminals, display managers, X clients, and window managers were successfully explored. The student gained practical understanding of Linux system interfaces and graphical session management.